

Table 52. Output voltage characteristics⁽¹⁾

Symbol	Parameter	Conditions	Min	Max	Unit
V_{OL}	Output low level voltage	CMOS port ⁽²⁾	-	0.4	V
V_{OH}	Output high level voltage	$ I_{IO} = 8 \text{ mA}$ $V_{DDIOx} \geq 2.7 \text{ V}$	$V_{DD} - 0.4$	-	V
$V_{OL}^{(3)}$	Output low level voltage	TTL port ⁽²⁾	-	0.4	V
$V_{OH}^{(3)}$	Output high level voltage	$ I_{IO} = 8 \text{ mA}$ $V_{DDIOx} \geq 2.7 \text{ V}$	2.4	-	V
$V_{OL}^{(3)}$	Output low level voltage	All I/Os	-	1.3	V
$V_{OH}^{(3)}$	Output high level voltage	$ I_{IO} = 20 \text{ mA}$ $V_{DDIOx} \geq 2.7 \text{ V}$	$V_{DD} - 1.3$	-	V
$V_{OL}^{(3)}$	Output low level voltage	$ I_{IO} = 4 \text{ mA}$	-	0.45	V
$V_{OH}^{(3)}$	Output high level voltage	$V_{DDIOx} \geq 1.65 \text{ V}$	$V_{DD} - 0.45$	-	V
$V_{OLFM+}^{(3)}$	Output low level voltage for an FT I/O pin in FM+ mode	$ I_{IO} = 20 \text{ mA}$ $V_{DDIOx} \geq 2.7 \text{ V}$	-	0.4	V
		$ I_{IO} = 10 \text{ mA}$ $V_{DDIOx} \geq 1.65 \text{ V}$	-	0.4	

1. The I_{IO} current sourced or sunk by the device must always respect the absolute maximum rating specified in [Table 21: Voltage characteristics](#). The sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings ΣI_{IO} .
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Specified by design. Not tested in production.

Input/output AC characteristics

The definition and values of input/output AC characteristics are given in [Figure 24](#) and [Table 53](#), respectively.

Unless otherwise specified, the parameters given are derived from tests performed under the ambient temperature and supply voltage conditions summarized in [Table 24: General operating conditions](#).

Table 53. I/O AC characteristics

Speed	Symbol	Parameter ⁽¹⁾⁽²⁾	Conditions	Min	Max	Unit
00	Fmax	Maximum frequency	$C=50 \text{ pF}, 2.7 \text{ V} \leq V_{DDIOx} \leq 3.6 \text{ V}$	-	2	MHz
			$C=50 \text{ pF}, 1.65 \text{ V} \leq V_{DDIOx} \leq 2.7 \text{ V}$	-	0.35	
			$C=10 \text{ pF}, 2.7 \text{ V} \leq V_{DDIOx} \leq 3.6 \text{ V}$	-	3.00	
			$C=10 \text{ pF}, 1.65 \text{ V} \leq V_{DDIOx} \leq 2.7 \text{ V}$	-	0.45	
	Tr/Tf	Output rise and fall time ⁽³⁾	$C=50 \text{ pF}, 2.7 \text{ V} \leq V_{DDIOx} \leq 3.6 \text{ V}$	-	100.00	ns
			$C=50 \text{ pF}, 1.65 \text{ V} \leq V_{DDIOx} \leq 2.7 \text{ V}$	-	225.00	
			$C=10 \text{ pF}, 2.7 \text{ V} \leq V_{DDIOx} \leq 3.6 \text{ V}$	-	75.00	
			$C=10 \text{ pF}, 1.65 \text{ V} \leq V_{DDIOx} \leq 2.7 \text{ V}$	-	150.00	